

MATH 1075 - Section 6.7 Worksheet

09/08/2026

Identifying Quadratic Equations

Determine if following equations are linear, quadratic, or neither

1. $4x^2 + 2x + 8 = 0$

2. $5 - x = 0$

3. $x^3 + 2x + 1 = 0$

4. $x(x - 1) = 0$

4. $x^2(x + 5) = 0$

Solving Quadratic Equations using Zero Product Rule

Solve following equations.

1. $(x - 4)(x + 3) = 0$

2. $(x - 1)(x + 12) = 0$

3. $(2x - 3)(3x + 2) = 0$

4. $(2x - 2)(5x + 4) = 0$

5. $x(7x + 2) = 0$

Solving Quadratic Equations using Factoring

Solve following equations using factoring.

1. $x^2 + 2x + 1 = 0$

2. $x^2 - 7x - 8 = 0$

3. $4x^2 - 11x = 3$

4. $9x^2 = 1$

Solving Higher Order Equations

Solve following equations.

1. $3x^3 - 6x^2 = 24x$

2. $5(x - 3)(4x + 1)(7x - 4) = 0$

3. $x^3 - 3x^2 - 4x + 12 = 0$